

Introduction to Computer Vision

Qingjie Meng

Assistant Professor
School of Computer Science
University of Birmingham



[34266][34256][LI Artificial Intelligence 2]

Lectures and tutorial

- Ata Kaban: week 1-6 (A.Kaban@bham.ac.uk, office hour: Monday 15:00-17:00)
- Qingjie Meng: week 7-8 (m.qingjie@bham.ac.uk, office hour (Wednesday 9:00-10:00)
CS 118)
- Miqing Li: week 9-11 (m.li.8@bham.ac.uk)



Two lectures per week: Friday morning 10:00-12:00
(14th and 21st November)
(Teaching & Learning Building, LG18)

Tutorial: Tuesday 15:00-16:00 (18th November)
(Teaching & Learning Building, LG18)

What is AI?

Google AI

Q All News Images Videos Books More Tools



Forbes
Where Does AI Go Next?



Plat.AI
Build an AI ...



Simplilearn
Artificial Intelligence ...



TIME
Using AI to Its Fullest Human Potential ...



Simplilearn
7 Types of Artificial Intelligence That ...



Insurtec
Insurtec



Emarsys
What Is Real Artificial Intelligence ...



Springboard
Is AI Hard To Learn? A Guide To Getting ...



Nature
Artificial Intelligence



The Hindu Business Line
Leveraging AI for tackling corruption ...



Camino Partners
AI Impact recruitment marketing ...



RAID Conf
AI's balancing



McKinsey
Artificial Intelligence Insights ...



gmailfoundry.ac.uk
Greater Manchester AI | Gre...



The World Economic Forum
A 5-step guide to scale responsible AI ...



The Times of Israel
All-in-one chips seen boosting computer ...



Corporate Finance Institute
Artificial Intelligence (AI) - Overview ...

What is AI?

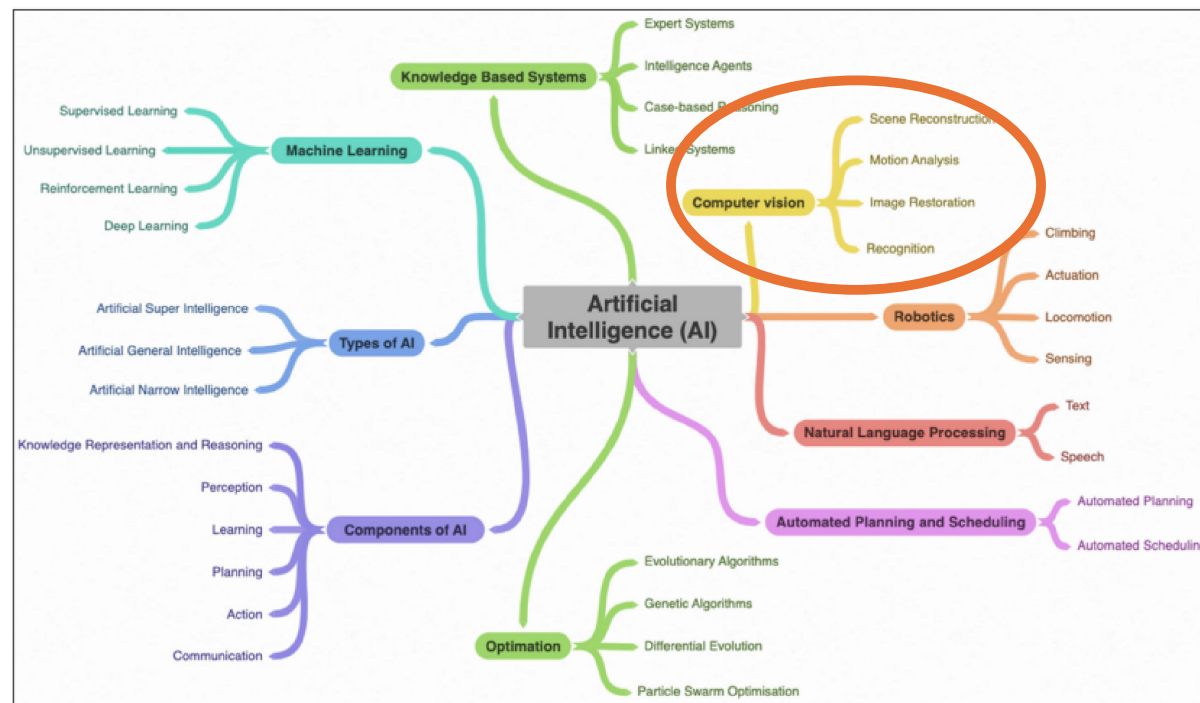


Figure: M. Regona et al., Opportunities and Adoption Challenges of AI in the Construction Industry: A PRISMA Review, JOITMC 8(1):45, 2022

- Lecture (Qingjie Meng)
 - PhD @ Imperial College London
 - PostDoc @ Imperial College London
 - Research area: AI for Medical Imaging
 - Teaching
 - Lecturer of Current Topics in AI / ML
 - Lecturer of Computer Vision and Imaging

Main agenda

- **Part 1 (Week 7)**
 - Introduction to computer vision
 - Deep learning
- **Part 2 (Week 8)**
 - Neural networks
 - Introduction to AI in medical imaging

Contents (Week 7)

- Introduction to computer vision
 - What is computer vision
 - Computer vision applications
 - Challenges in computer vision
 - Why vision
- Deep learning
 - Introduction
 - Classic deep neural networks
 - Implementation

What is computer vision?

- Computing properties of the 3D world from visual data (*measurement*)

Real-time stereo



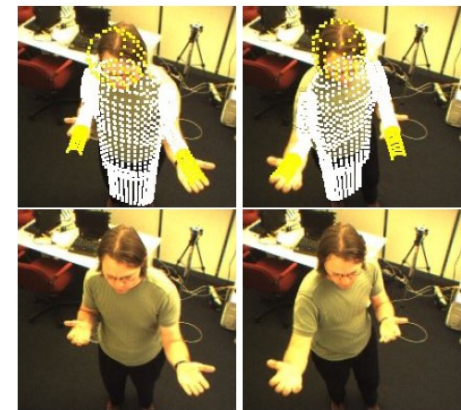
Wang et al.

Structure from motion



Snaveley et al.

Tracking

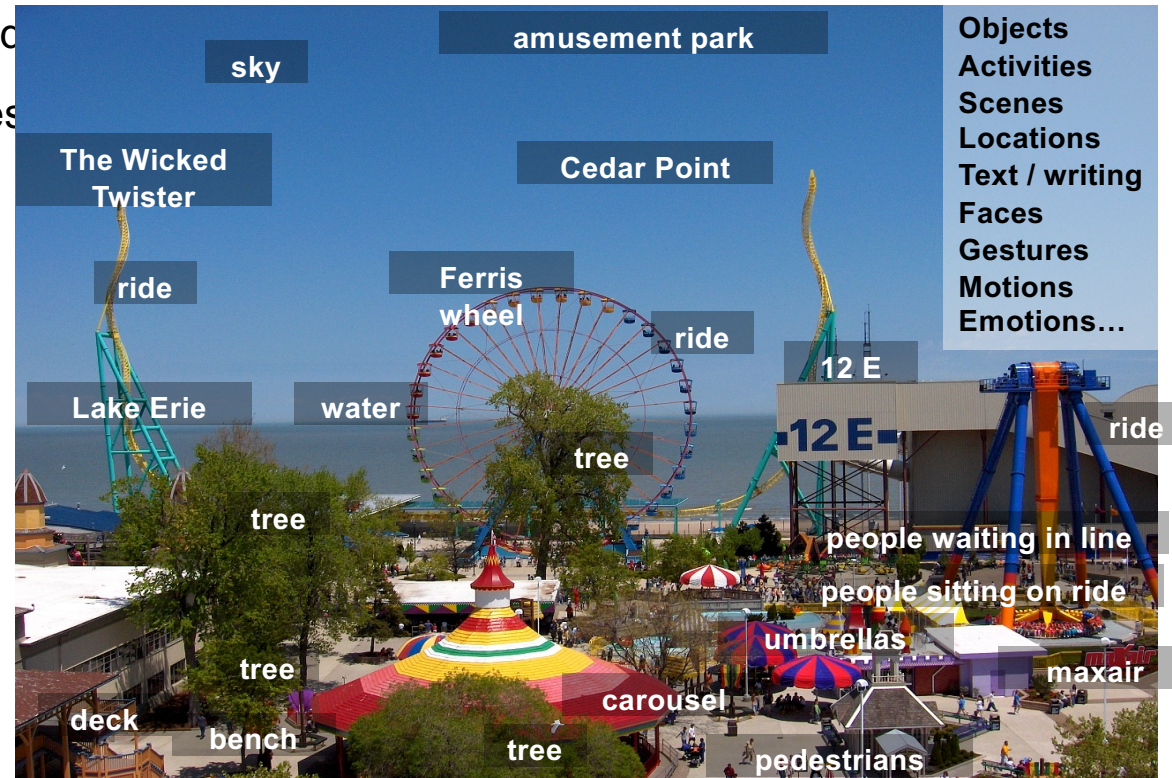


Demirdjian et al.

Slide credit: Kristen Grauman

What is computer vision?

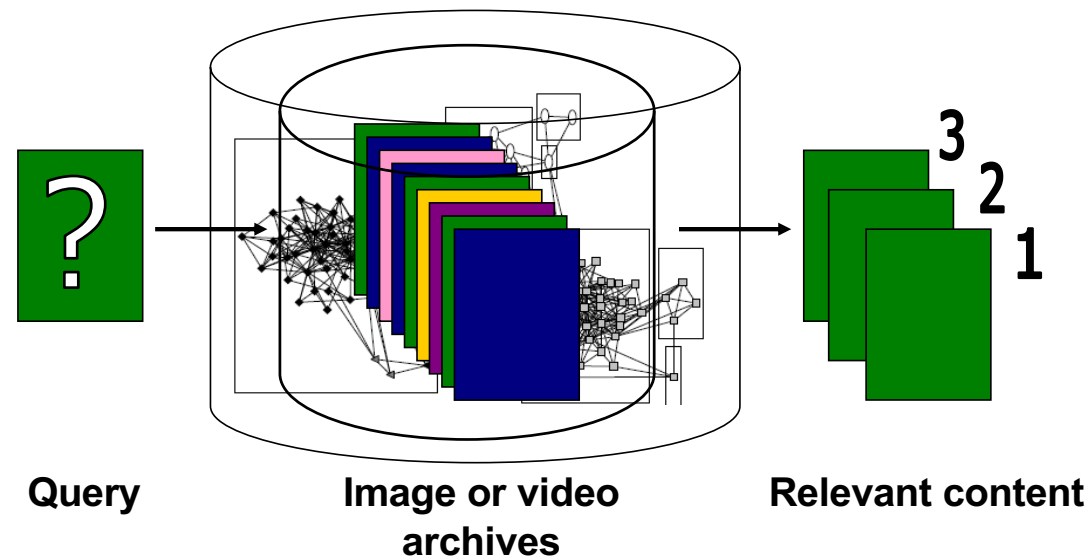
- Algorithms and
people, scenes



Slide credit: Kristen Grauman

What is computer vision?

- Algorithms to mine, search, and interact with visual data
- (*search and organisation*)

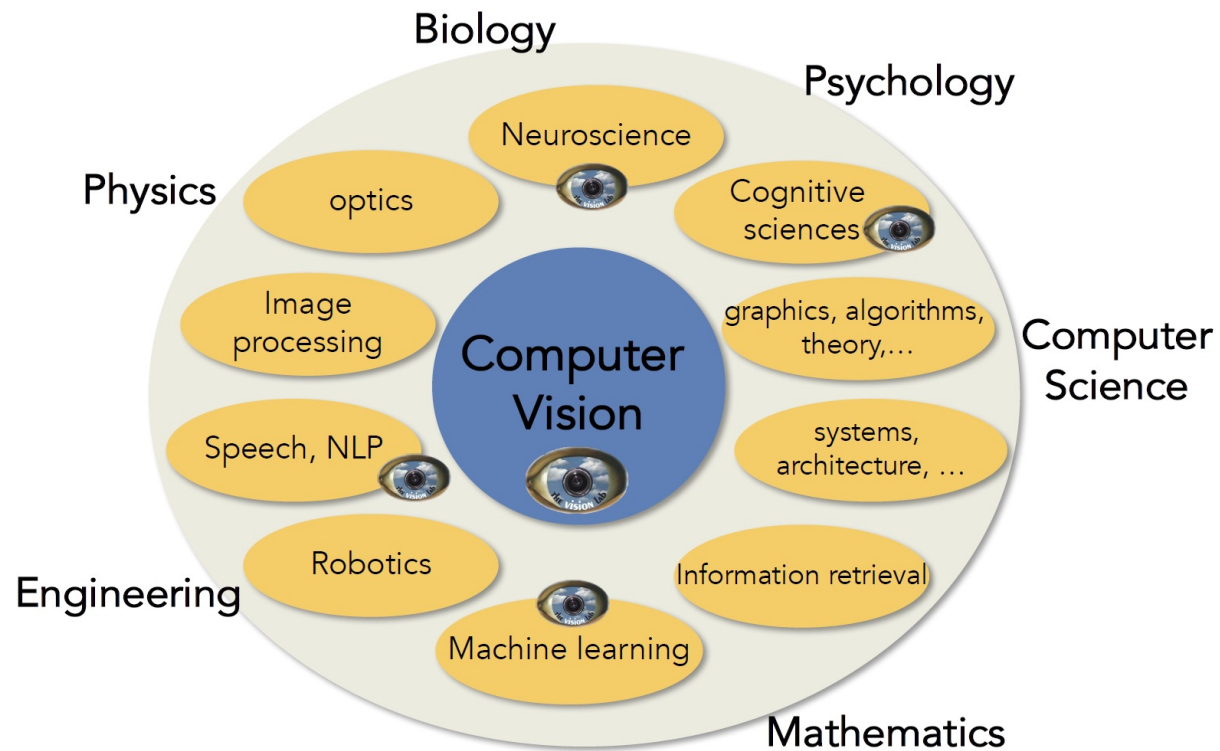


Slide credit: Kristen Grauman

What is computer vision?

- Reconstructing 2D/3D visual environments and objects
(visualisation)

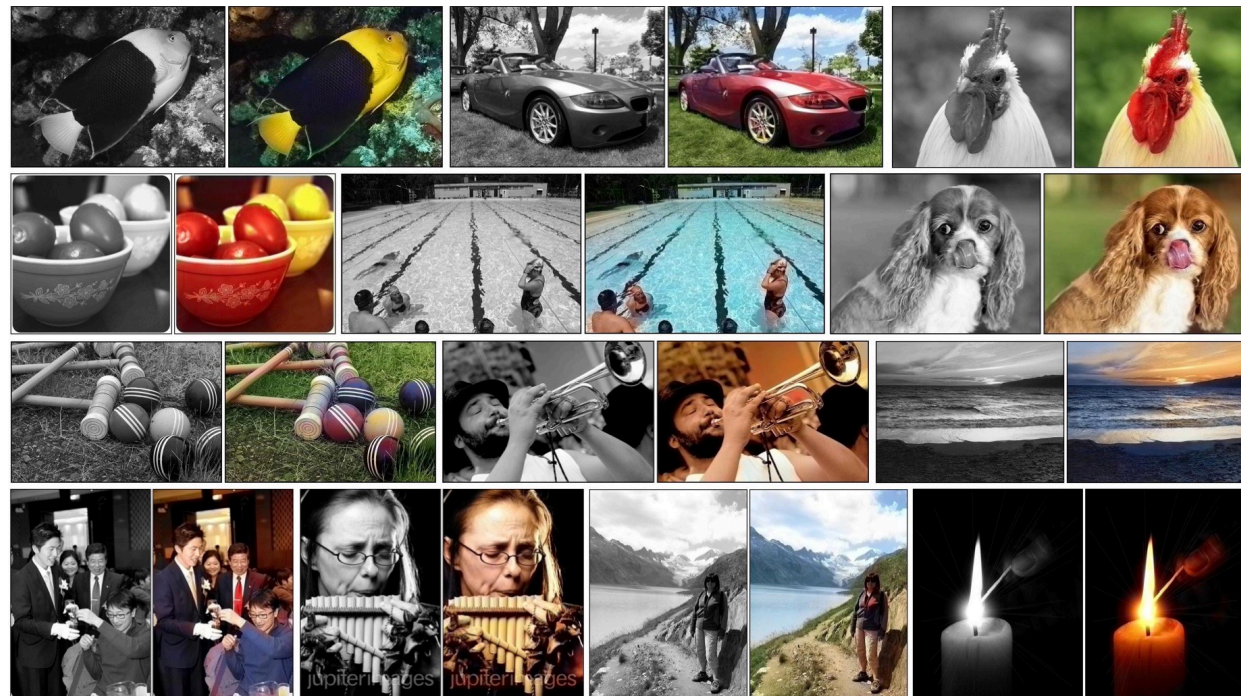
What is computer vision?



Slide credit: Fei-Fei Li

Computer vision applications

Image colourisation



Computer vision applications

App Store Preview

Open the Mac App Store to buy and download apps.

Colorize - Color to Old Photos 

Black and White Colorization
Photomyne LTD
Designed for iPhone

★★★★★ 4.7 • 6.9K Ratings

Free - Offers in-App Purchases

[View in Mac App Store](#)

iPhone Screenshots

Before & After

Colorize Black-and-White photos

Before After

See the magic as color is added

Brought to you by the developer that built the smartest photo scanner app, Photomyne is proud to now also offer an AI-powered B&W colorization app that will blow you away. It's one of the most accurate old photo colorization apps - the results will amaze you.

IT'S AUTOMATIC - JUST ADD B&W PHOTOS: [more](#)

Google Play Games Apps Movies & TV Books Kids

Colorize Images

Colorize Images
Contains ads. In-app purchases

3.9★ 13.6K reviews 1M+ Downloads Everyone

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Developer contact

Similar apps

- Colorize - Color to Old Photos
Photomyne Ltd.
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Vertexshare Software Ltd
4.4★
- Print adjuster
krem151
4.7★
- Sketch on PDF
Applicationsprogramvara
3.0★

About this app

Bring new life to your old black and white photos with Colorize Images. Our AI-powered app uses advanced machine learning algorithms to automatically add vibrant colors to your grayscale images, giving them a fresh and modern look.

With a simple and intuitive interface, Colorize Images makes it easy to transform your photos into colorful works of art. Select a photo from your device's gallery or share it from another app, and let our app do the rest. Save your colorized photos to your gallery, or share them with friends and family on social media.

Park, 1941

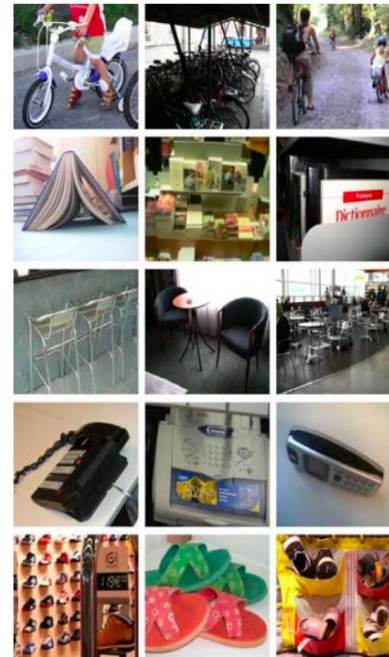
Textile Mill, June 1937

Berry Field, June 1909

Hamilton, 1936

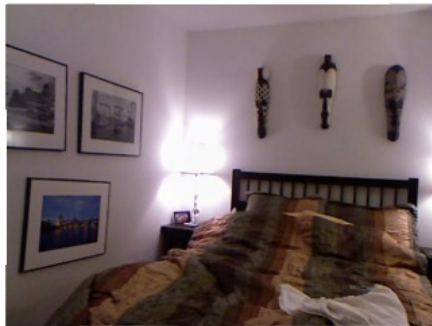
Computer vision applications

Image classification



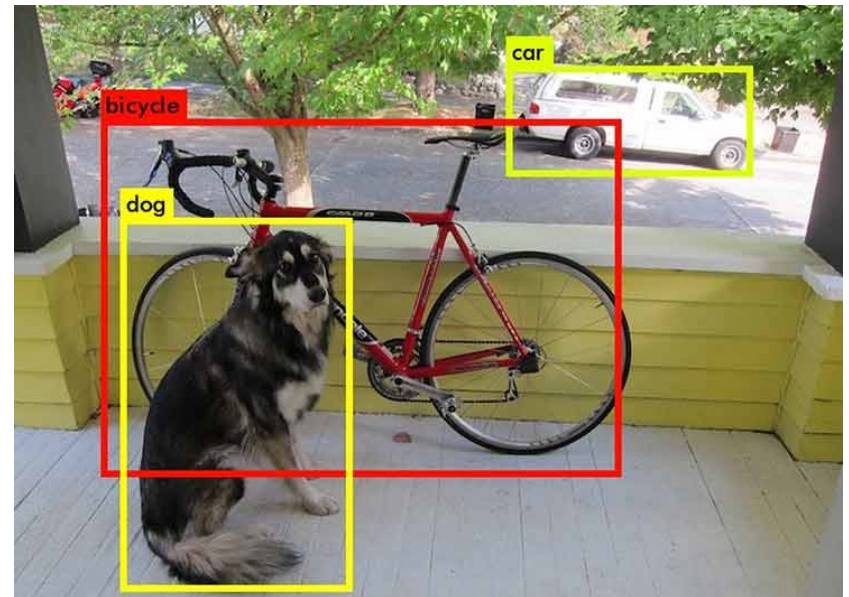
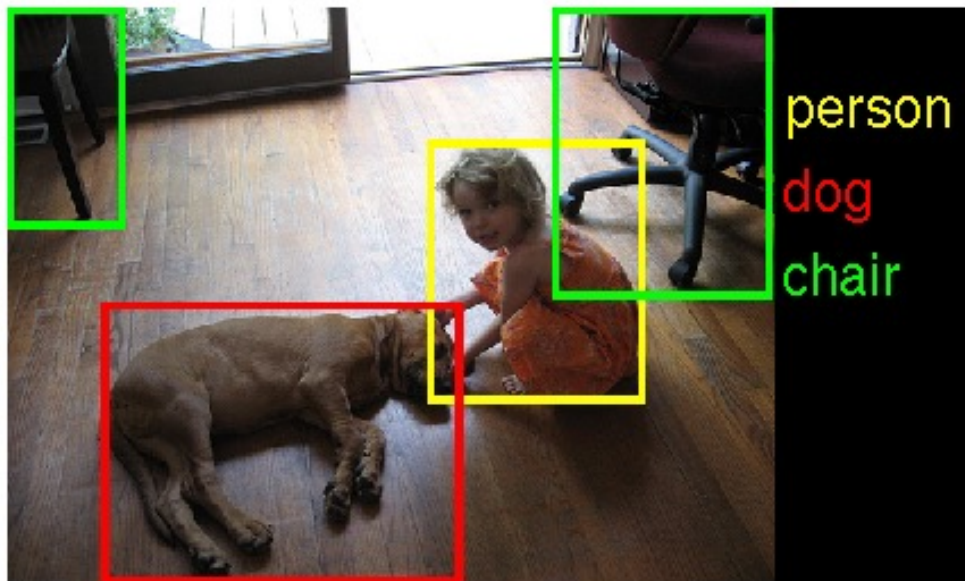
Computer vision applications

Image segmentation



Computer vision applications

Object detection



Con

The latest version YOLOv11

YOLO v2

<http://pureddie.com/yolo>

Computer vision applications



Computer vision applications

Face recognition



Computer vision applications

Face recognition



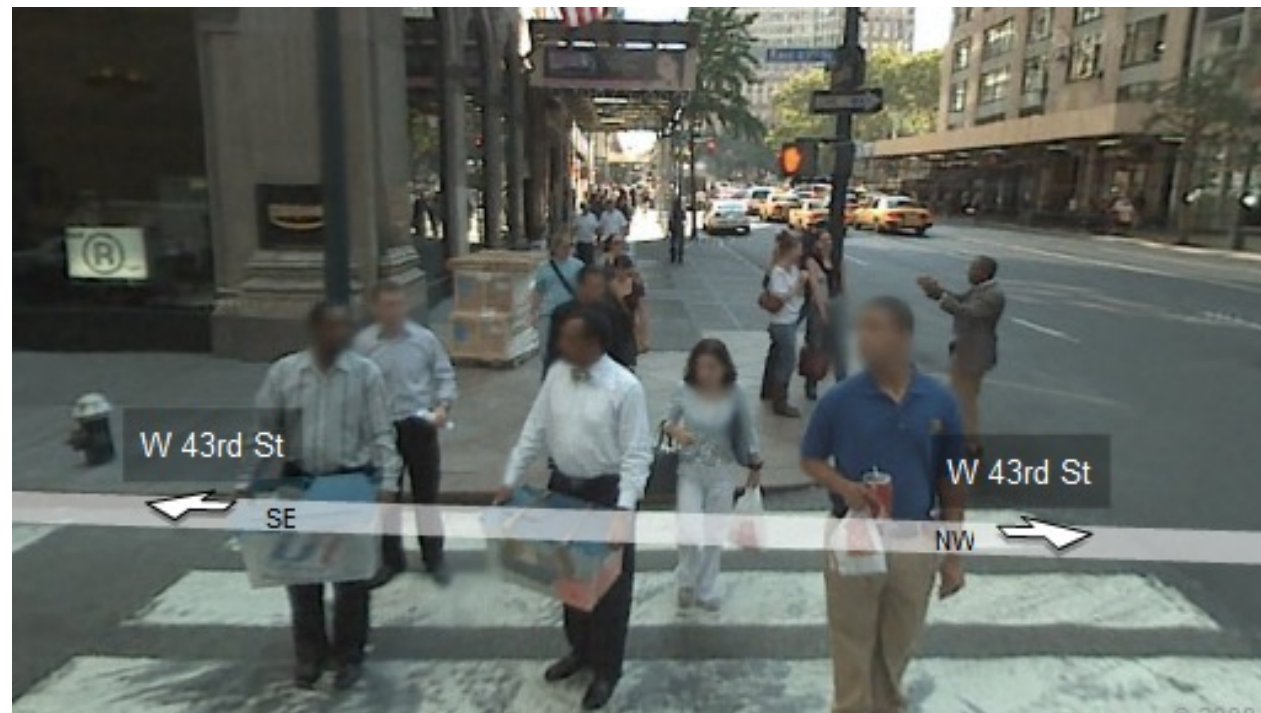
**HOW FACE ID
ACTUALLY WORKS**

Computer vision applications



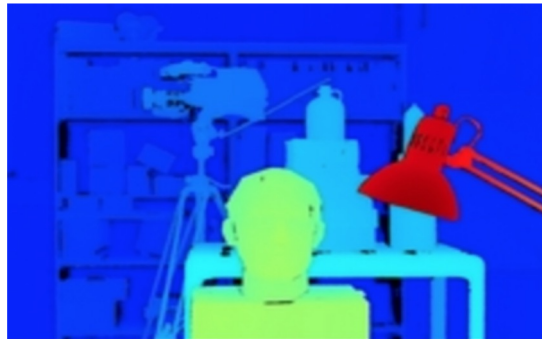
Computer vision applications

Face recognition for privacy protection



Computer vision applications

Depth estimation



Computer vision applications

3D reconstruction from photo collections

Colosseum, Rome, Italy



San Marco Square, Venice, Italy



Q. Shan, R. Adams, B. Curless, Y. Furukawa, and S. Seitz,
[The Visual Turing Test for Scene Reconstruction](#), 3DV
2013

Slide credit: Svetlana Lazebnik

Computer vision applications

3D rendering



<https://datagen.tech/guides/synthetic-data/neural-radiance-field-nerf/>

<https://waymo.com/research/block-nerf/>

Co



<https://www.youtube.com/watch?v=DJ2hcC1orc4>

Cor

PhysGaussian: Physics-Integrated 3D Gaussians for Generative Dynamics

Tianyi Xie^{1*}, Zeshun Zong^{1*}, Yuxing Qiu^{1*}, Xuan Li^{1*},
Yutao Feng^{2,3}, Yin Yang³, Chenfanfu Jiang¹

UCLA



浙江大学
ZHEJIANG UNIVERSITY

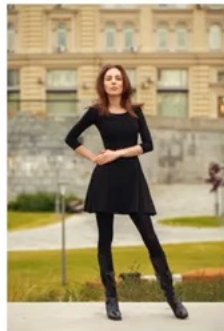
THE UNIVERSITY
OF UTAH

* Equal contributions

[with audio 🗣️]

Computer vision applications

Human digitization



Computer vision applications

Human digitization



Computer vision applications

Augmented reality

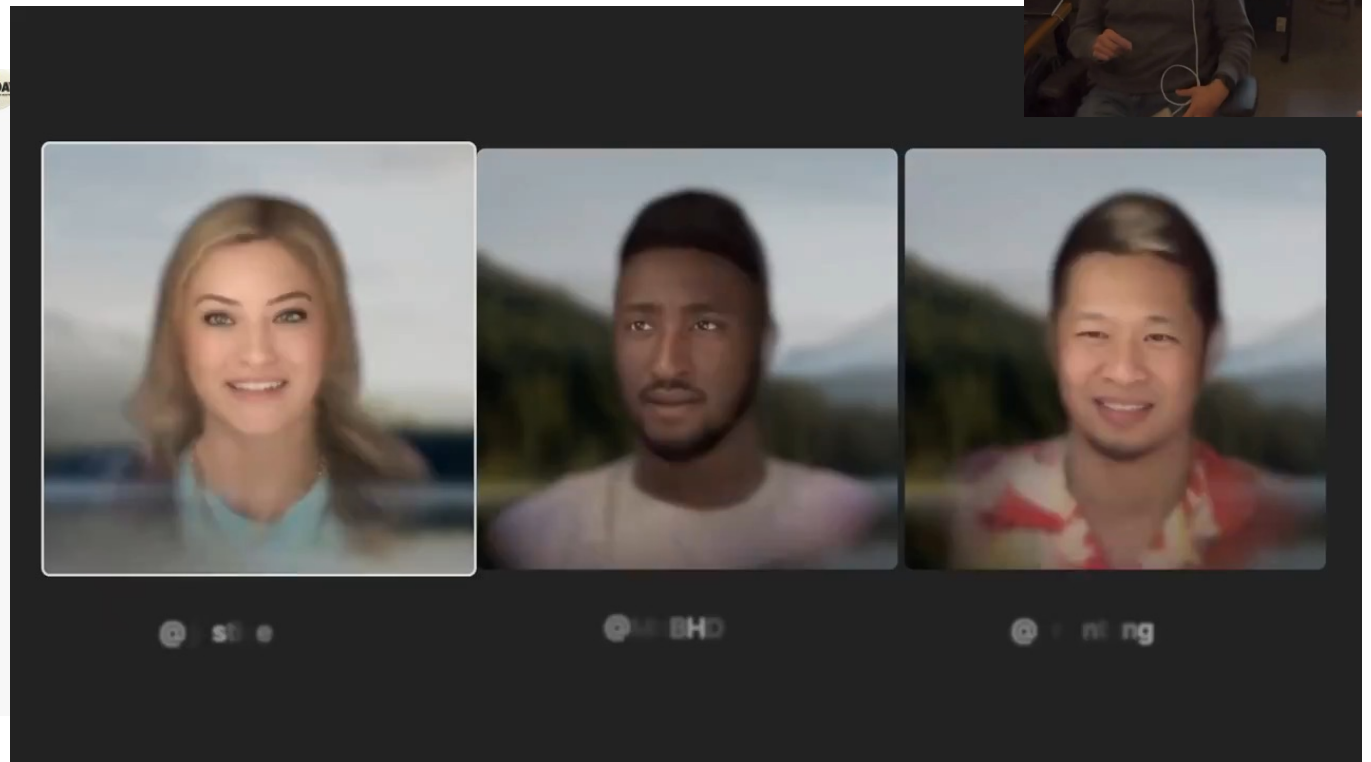


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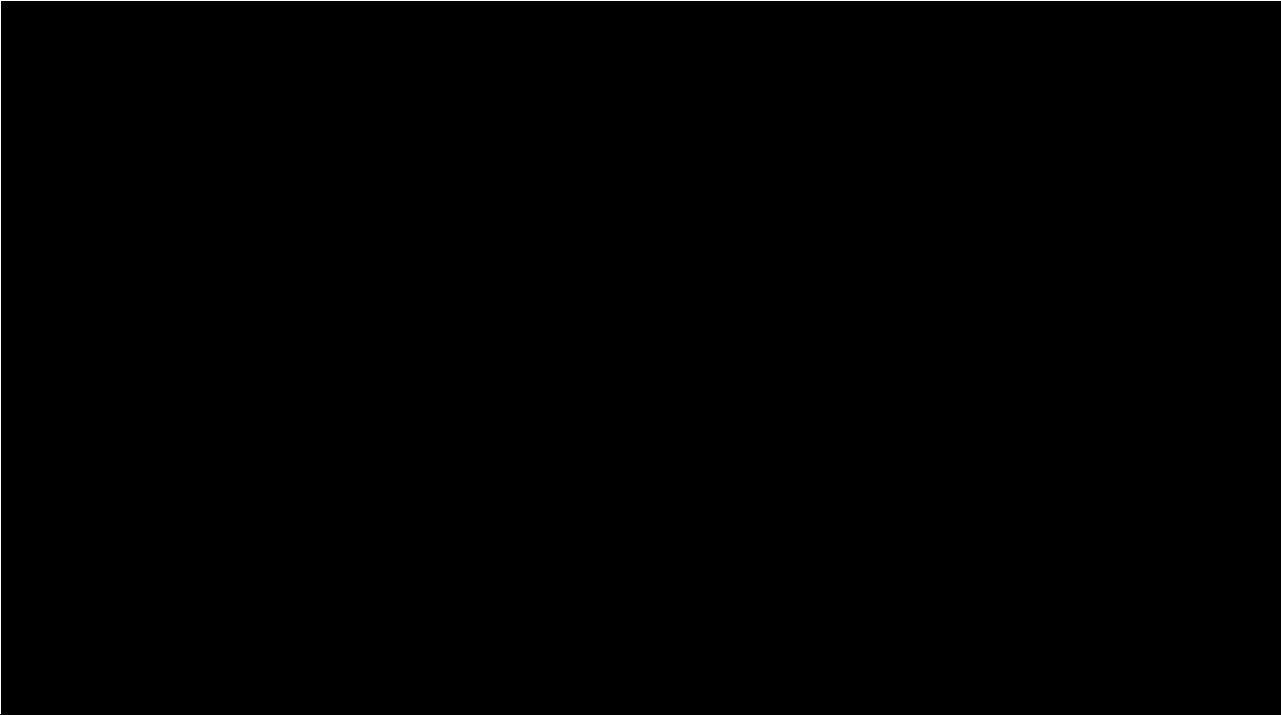
Computer vision applications

Apple Vision Pro



Computer vision applications

Autonomous Vehicle



Computer vision applications

OpenAI DALL·E

Image generation



Computer vision applications

OpenAI Sora

Video generation



Challenges in computer vision

Ambiguity



Slide credit: Svetlana Lazebnik

Challenges in computer vision

Many nuisance parameters



Illumination



Object pose



Clutter



Viewpoint

Slide credit: Kristen Grauman

Challenges in computer vision

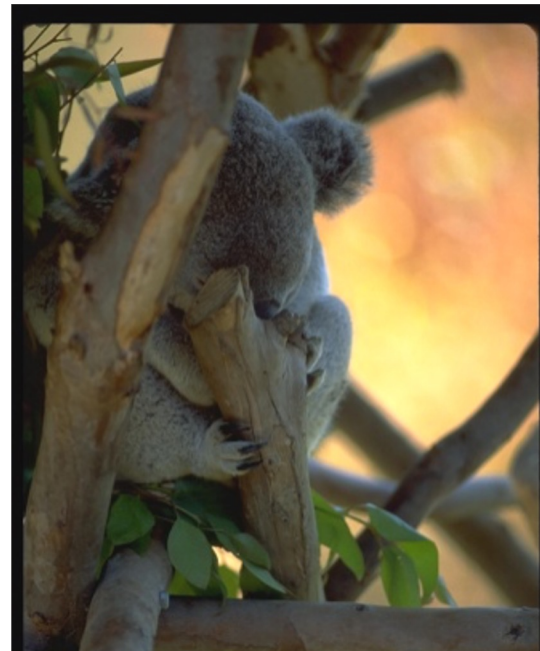
Motion



Slide credit: Svetlana Lazebnik

Challenges in computer vision

Occlusion



Slide credit: Svetlana Lazebnik

Challenges in computer vision

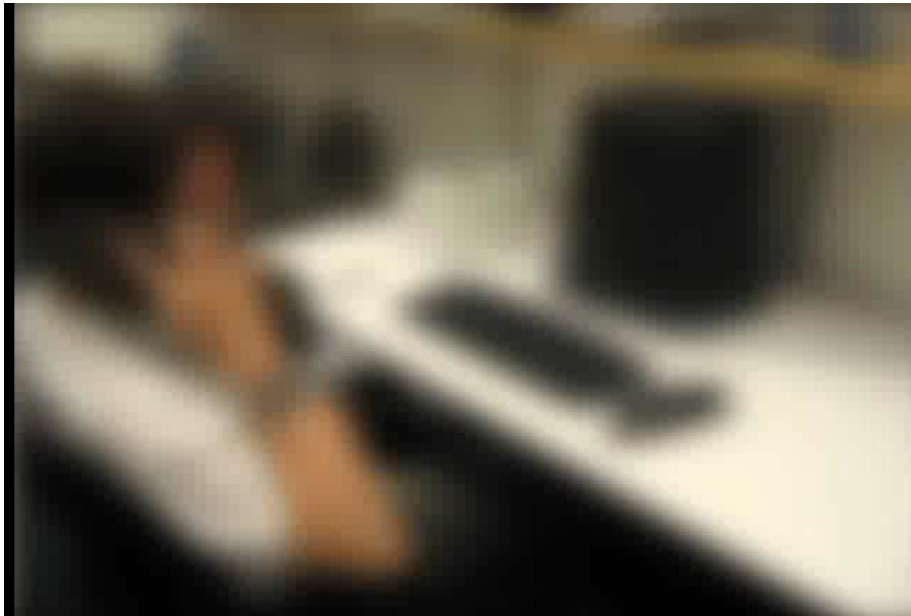
Object intra-class variation



Slide credit: Fei-Fei, Fergus, Torralba

Challenges in computer vision

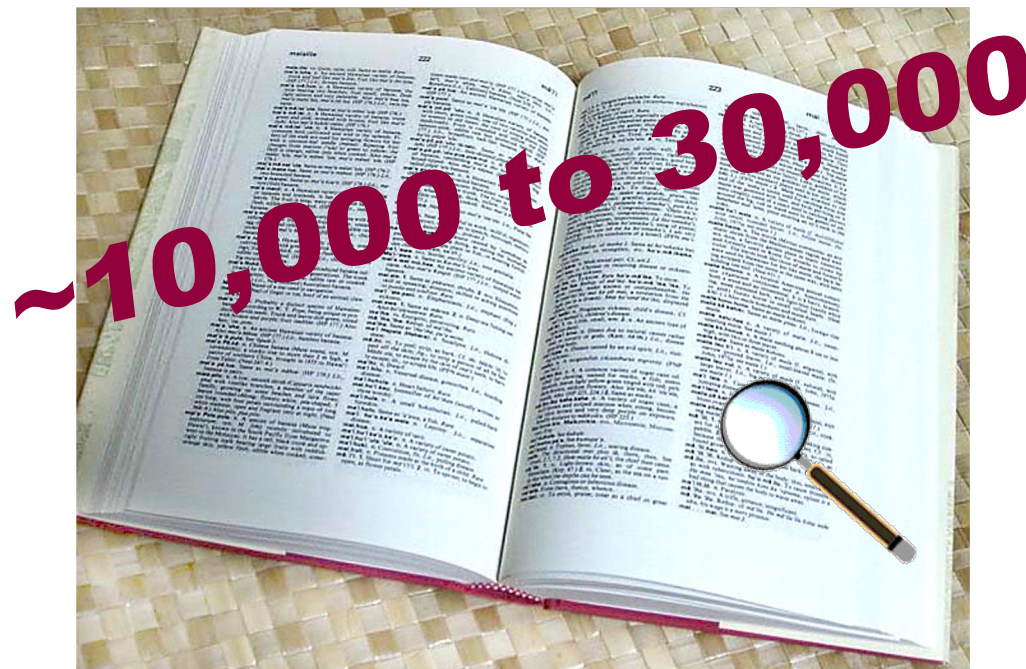
Context and human experience



Slide credit: Fei-Fei, Fergus, Torralba

Challenges in computer vision

Complexity



Slide credit: Fei-Fei, Fergus, Torralba

Challenges in computer vision

Complexity

The Flickr logo, with the word "flickr" in a blue and pink sans-serif font.

10 billion images

The Facebook logo, consisting of the word "facebook" in white lowercase letters on a blue rectangular background.

250 billion images

The imgur logo, featuring the text "the simple image sharer" in small green letters above the word "imgur" in a large, bold, black sans-serif font.

1 billion images served daily

The YouTube logo, with the word "You" in black and "Tube" in white inside a red rounded rectangle.

500 hours uploaded per minute

The Photobucket logo, featuring a blue camera icon followed by the word "photobucket" in a blue sans-serif font.

10 billion images

From Cisco

The Cisco logo, which consists of a stylized bridge icon made of vertical lines of varying heights, followed by the word "CISCO" in a red sans-serif font.

Almost **90%** of web traffic is visual!

Challenges in computer vision

Complexity

- Thousands to millions of pixels in an image
- 30+ degrees of freedom in the pose of articulated objects (humans)

Why vision?

- As image sources multiply, so do applications
 - Relieve humans of boring, easy tasks
 - Enhance human abilities
 - Advance human-computer interaction, visualisation
 - Perception for robotics / autonomous agents
 - Organise and give access to visual content

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Top conferences



Journals

IEEE Transactions on Pattern Analysis and Machine Intelligence

18.6
Impact Factor



International Journal of Computer Vision

Publishing model
Hybrid

[Browse Journals & Magazines](#) > [IEEE Transactions on Image Processing](#) 

IEEE Transactions on Image Processing

13.7
Impact Factor

0.08566
Eigenfactor

3.736
Article Influence Score

22.5
CiteScore
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Innovative Applications of Computer Vision

- Verifying financial documentation
- Assessing the health of ocean ecosystem
- Streamlining supply chain operations
- Optimizing Retail Inventory Management And Marketing
- Automating QA For Manufacturers
- Automating Retail Checkout
- Creating Knowledge Bases
- Converting Paper Documents Into Digital Files
- Spotting Phishing Content
- Predicting Potential Security Threats
- Assessing The Appeal Of Real Estate
- Monitoring The Health Of Crops
- Enabling Visual Search
- Conserving Wildlife And Natural Resources
- Making Workplaces Safer
- Monitoring Green Energy Generation
- Hyper-Personalizing Shopping Selections
- Analyzing Medical Images
- Screening Job Candidates
- Powering Immersive AR

<https://www.forbes.com/sites/forbestechcouncil/2024/02/16/take-a-look-at-innovative-applications-of-computer-vision/amp/>

Related modules @UoB

- 30213, 30241 Computational Vision & Imaging
- 25024 Robot Vision
- 32167 Neural Computation
- 30227 Intelligent Robotics
- 41479/41482 Current Topics in DS
- 34149/32257 Current Topics in AI/ML
- 37864/35447 Programming for Data Science
- 40414/34147 AI Programming

Topics covered in CV&I / Robot vision

- Introduction to computer vision & human vision
- Image formation and colour
- Camera parameters
- Camera calibration
- Image noise
- Image filtering
- Image gradient
- Edge detection
- Matching
- Binary image analysis
- Image texture
- Image segmentation
- Fitting: voting and the Hough transform
- Image warping
- Image Stitching
- Local features: image matching and detection
- Multiple-view Geometry
- Active Imaging (3D imaging technologies)
- Face recognition
- Deep learning for computer vision I
- Deep learning for computer vision II

Summary

- The core of AI lies in computer vision, which has recently driven its drastic success.
- It is a fascinating and challenging field, with plenty of useful and surprising applications.
- Computer vision has opened up new opportunities, with the emergence of new companies, startups, and more.

Q & A